PS-020

Local SRv6 Path Steering and Measured Failure Recovery

AIORI-3
6G & Future Networks

AIORI-3 | Category: 6G & Future Networks

Team: [Team Name] | Team Code: [A3-PS020-TXXXX]

Project focus: compare explicit SRv6 paths with ordinary hop-by-hop IPv6 routing

Objective: measure packet loss, interruption, detection delay, repair delay, and header overhead

Abstract & Problem Description

Traditional IP routing forwards packets hop-by-hop using paths calculated by routing protocols.

SRv6 allows an ingress router to steer a packet along an explicit path using an ordered list of Segment Identifiers (SIDs).

The challenge is to observe what happens before, during, and after a controlled link failure.

The project compares both approaches on the same five-router topology and failure scenario.

Background: SRv6 Path Steering

SRv6 means Segment Routing over IPv6.

An ordered list of SIDs is encoded in the IPv6 packet header to specify an explicit path.

The testbed contains five virtual IPv6/SRv6 routers.

Two explicit SRv6 paths will be configured, alongside an ordinary IPv6 routing baseline.

Literature & RFC Review

Primary reference: RFC 8754 — Segment Routing over IPv6 (SRv6).

The problem statement requires packet captures to verify correct SID traversal.

The comparison must use an identical topology and controlled failure scenario.

No guaranteed operational advantage is assumed; results must come from measurements.

Proposed Solution

1. Build a five-router virtual IPv6/SRv6 testbed.
2. Configure two explicit SRv6 SID paths and an ordinary IPv6 routing baseline.
3. Generate traffic and capture packets to verify SID traversal.
4. Introduce one controlled link failure.
5. Use a script to switch to a backup policy.
6. Repeat the failure scenarios and compare the measurements.

Solution Optimization: What We Will Measure

Packet loss: packets not received during the test.

Interruption time: duration of the traffic disruption.

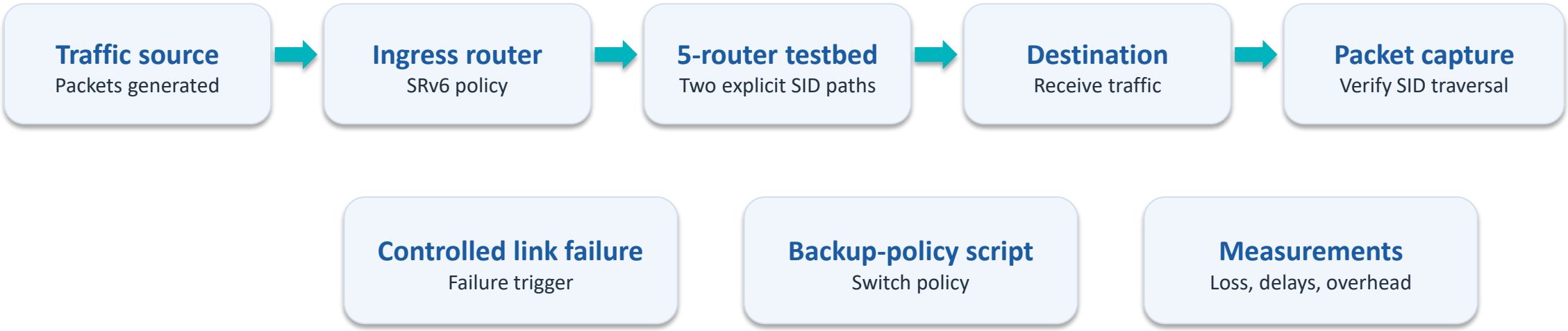
Failure detection delay: time taken to detect the failure.

Repair delay: time taken for traffic to recover after the failure.

SRv6 header overhead: additional header cost associated with the explicit path.

Repeat scenarios to improve measurement reliability.

Solution Architecture



Traffic source → ingress router → selected path through five-router testbed → destination

Packet capture verifies SID traversal.

Link-failure trigger activates the scripted backup-policy switch.

Measurements are collected for both SRv6 and ordinary IPv6 routing.

Data Flow & Failure-Recovery Workflow

1. Baseline
Generate traffic

2. Failure
Break one link

3. Observe
Capture loss and interruption

4. Recover
Switch backup policy

5. Measure
Detection and repair delay

6. Compare
Repeat for both routing modes

Start baseline traffic and record normal operation.

Trigger a controlled link failure.

Record packet loss and interruption.

Detect failure and switch to the backup policy.

Record repair delay and capture packets.

Repeat, then compare SRv6 with ordinary IPv6 routing.

Deployment Timeline & Validation

Phase 1 — Set up the private GitHub repository using the team code.

Phase 2 — Add aiori-hackathon as collaborator and transfer ownership after acceptance.

Phase 3 — Build the topology, configure paths, and prepare the failure script.

Phase 4 — Capture packets, repeat tests, and analyse measurements.

Phase 5 — Upload the PDF presentation, source code, supporting files, and demo link/files.

AIORI timeline: Level 1 steps due 14 Oct; Level 2 announcement 16 Oct; mentoring 16–31 Oct.

Expected Deliverables

Five-router SRv6 topology and path-steering configuration

Ordinary IPv6 routing baseline

Scripted failure and backup-policy mechanism

Packet captures demonstrating SID traversal

Repeated-failure performance comparison

Analysis of loss, recovery, and header overhead

Final technical report and prototype demonstration

Prototype demonstration with a pseudocode snippet to the mentor

References & Submission Checklist

RFC 8754: https://www.rfc-editor.org/rfc/rfc8754.html

PS-020: Local SRv6 Path Steering and Measured Failure Recovery (AIORI-3 problem statement)

Use the proposed structure: Abstract, Problem Description, Background, RFC Review, Proposed Solution, Optimization, Architecture, Data Flow, Timeline, Bibliography.

Upload the presentation as PDF; add the YouTube video link to the team GitHub repository.

Follow all ten AIORI guideline steps; the document states that missing required steps can lead to elimination.

Thank You

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